Integration by Substitution

Step-by-Step Worked Solutions

Introductory problems 10(a)–(b) and Exercises 15–56

Section 1: Introductory Problems

10(a) $\displaystyle\int \frac{\sin 3\theta}{1+\cos 3\theta}\,d\theta$

Substitution: Let $u = 1+\cos 3\theta$, then $du = -3\sin 3\theta\,d\theta \implies \sin 3\theta\,d\theta = -\frac{1}{3}\,du$.

Evaluation: $-\frac{1}{3}\displaystyle\int \frac{du}{u} = \boxed{-\frac{1}{3}\ln|1+\cos 3\theta| + C}$

10(b) $\displaystyle\int \frac{e^x}{1+e^x}\,dx$

Substitution: Let $u = 1+e^x$, then $du = e^x\,dx$.

Evaluation: $\displaystyle\int \frac{du}{u} = \ln|1+e^x| + C = \boxed{\ln(1+e^x) + C}$

Section 2: Exercises 15–42

Ex.	Integral	Choice of u and du	Transformed integral & final answer
15	$\displaystyle\int (4x-3)^9\,dx$	$u = 4x-3 \implies dx = \frac{du}{4}$	$\frac{1}{4}\displaystyle\int u^9\,du$ $= \boxed{\frac{1}{40}(4x-3)^{10} + C}$
16	$\displaystyle\int x^3\sqrt{5+x^4}\,dx$	$u = 5+x^4 \implies x^3dx = \frac{du}{4}$	$\frac{1}{4}\displaystyle\int u^{1/2}\,du$ $= \boxed{\frac{1}{6}(5+x^4)^{3/2} + C}$
17	$\displaystyle\int \sin 7x\,dx$	$u = 7x \implies dx = \frac{du}{7}$	$\frac{1}{7}\displaystyle\int \sin u\,du$ $= \boxed{-\frac{1}{7}\cos 7x + C}$

Ex.	Integral	Choice of u and du	Transformed integral & final answer
18	$\int \cos\left(\frac{x}{3}\right) dx$	$u = \frac{x}{3} \implies dx = 3\,du$	$3\int \cos u\,du$ $= \boxed{3\sin\left(\frac{x}{3}\right) + C}$
19	$\int \sec 4x \tan 4x\,dx$	$u = 4x \implies dx = \frac{du}{4}$	$\frac{1}{4}\int \sec u \tan u\,du$ $= \boxed{\frac{1}{4}\sec 4x + C}$
20	$\int \sec^2 5x\,dx$	$u = 5x \implies dx = \frac{du}{5}$	$\frac{1}{5}\int \sec^2 u\,du$ $= \boxed{\frac{1}{5}\tan 5x + C}$
21	$\int e^{2x}\,dx$	$u = 2x \implies dx = \frac{du}{2}$	$\frac{1}{2}\int e^u\,du$ $= \boxed{\frac{1}{2}e^{2x} + C}$
22	$\int \frac{dx}{2x}$	$u = 2x \implies dx = \frac{du}{2}$	$\frac{1}{2}\int \frac{du}{u}$ $= \boxed{\frac{1}{2}\ln\lvert x\rvert + C}$
23	$\int \frac{dx}{\sqrt{1-4x^2}}$	$u = 2x \implies dx = \frac{du}{2}$	$\frac{1}{2}\int \frac{du}{\sqrt{1-u^2}}$ $= \boxed{\frac{1}{2}\arcsin(2x) + C}$
24	$\int \frac{dx}{1+16x^2}$	$u = 4x \implies dx = \frac{du}{4}$	$\frac{1}{4}\int \frac{du}{1+u^2}$ $= \boxed{\frac{1}{4}\arctan(4x) + C}$
25	$\int t\sqrt{7t^2+12}\,dt$	$u = 7t^2 + 12 \implies t\,dt = \frac{du}{14}$	$\frac{1}{14}\int u^{1/2}\,du$ $= \boxed{\frac{1}{21}(7t^2+12)^{3/2} + C}$
26	$\int \frac{x}{\sqrt{4-5x^2}}\,dx$	$u = 4 - 5x^2 \implies x\,dx = -\frac{du}{10}$	$-\frac{1}{10}\int u^{-1/2}\,du$ $= \boxed{-\frac{1}{5}\sqrt{4-5x^2} + C}$

Ex.	Integral	Choice of u and du	Transformed integral & final answer
27	$\int \frac{6}{(1-2x)^3}\,dx$	$u = 1-2x \implies dx = -\frac{du}{2}$	$-3\int u^{-3}\,du$ $= \boxed{\frac{3}{2(1-2x)^2} + C}$
28	$\int \frac{x^2+1}{\sqrt{x^3+3x}}\,dx$	$u = x^3+3x \implies (x^2+1)dx = \frac{du}{3}$	$\frac{1}{3}\int u^{-1/2}\,du$ $= \boxed{\frac{2}{3}\sqrt{x^3+3x} + C}$
29	$\int \frac{x^3}{(5x^4+2)^3}\,dx$	$u = 5x^4+2 \implies x^3dx = \frac{du}{20}$	$\frac{1}{20}\int u^{-3}\,du$ $= \boxed{-\frac{1}{40(5x^4+2)^2} + C}$
30	$\int \frac{\sin(1/x)}{3x^2}\,dx$	$u = \frac{1}{x} \implies \frac{dx}{x^2} = -du$	$-\frac{1}{3}\int \sin u\,du$ $= \boxed{\frac{1}{3}\cos\left(\frac{1}{x}\right) + C}$
31	$\int e^{\sin x}\cos x\,dx$	$u = \sin x \implies du = \cos x\,dx$	$\int e^u\,du$ $= \boxed{e^{\sin x} + C}$
32	$\int x^3 e^{x^4}\,dx$	$u = x^4 \implies x^3dx = \frac{du}{4}$	$\frac{1}{4}\int e^u\,du$ $= \boxed{\frac{1}{4}e^{x^4} + C}$
33	$\int x^2 e^{-2x^3}\,dx$	$u = -2x^3 \implies x^2dx = -\frac{du}{6}$	$-\frac{1}{6}\int e^u\,du$ $= \boxed{-\frac{1}{6}e^{-2x^3} + C}$
34	$\int \frac{e^x+e^{-x}}{e^x-e^{-x}}\,dx$	$u = e^x-e^{-x} \implies du = (e^x+e^{-x})dx$	$\int \frac{du}{u}$ $= \boxed{\ln\lvert e^x-e^{-x}\rvert + C}$
35	$\int \frac{e^x}{1+e^{2x}}\,dx$	$u = e^x \implies du = e^x\,dx$	$\int \frac{du}{1+u^2}$ $= \boxed{\arctan(e^x) + C}$
36	$\int \frac{t}{t^4+1}\,dt$	$u = t^2 \implies t\,dt = \frac{du}{2}$	$\frac{1}{2}\int \frac{du}{u^2+1}$ $= \boxed{\frac{1}{2}\arctan(t^2) + C}$

Ex.	Integral	Choice of u and du	Transformed integral & final answer
37	$\displaystyle\int \frac{\sin(5/x)}{x^2}\,dx$	$u = \dfrac{5}{x} \implies \dfrac{dx}{x^2} = -\dfrac{du}{5}$	$-\dfrac{1}{5}\displaystyle\int \sin u\,du$ $= \boxed{\dfrac{1}{5}\cos\left(\dfrac{5}{x}\right) + C}$
38	$\displaystyle\int \frac{\sec^2(\sqrt{x})}{\sqrt{x}}\,dx$	$u = \sqrt{x} \implies \dfrac{dx}{\sqrt{x}} = 2\,du$	$2\displaystyle\int \sec^2 u\,du$ $= \boxed{2\tan(\sqrt{x}) + C}$
39	$\displaystyle\int \cos^4 3t\,\sin 3t\,dt$	$u = \cos 3t \implies \sin 3t\,dt = -\dfrac{du}{3}$	$-\dfrac{1}{3}\displaystyle\int u^4\,du$ $= \boxed{-\dfrac{1}{15}\cos^5 3t + C}$
40	$\displaystyle\int \cos 2t\,\sin^5 2t\,dt$	$u = \sin 2t \implies \cos 2t\,dt = \dfrac{du}{2}$	$\dfrac{1}{2}\displaystyle\int u^5\,du$ $= \boxed{\dfrac{1}{12}\sin^6 2t + C}$
41	$\displaystyle\int x\sec^2(x^2)\,dx$	$u = x^2 \implies x\,dx = \dfrac{du}{2}$	$\dfrac{1}{2}\displaystyle\int \sec^2 u\,du$ $= \boxed{\dfrac{1}{2}\tan(x^2) + C}$
42	$\displaystyle\int \frac{\cos 4\theta}{(1+2\sin 4\theta)^4}\,d\theta$	$u = 1 + 2\sin 4\theta \implies \cos 4\theta\,d\theta = \dfrac{du}{8}$	$\dfrac{1}{8}\displaystyle\int u^{-4}\,du$ $= \boxed{-\dfrac{1}{24(1+2\sin 4\theta)^3} + C}$

Section 3: Exercises 43–56

Ex.	Integral	Choice of u and du	Transformed integral & final answer
43	$\displaystyle\int \cos 4\theta\sqrt{2-\sin 4\theta}\,d\theta$	$u = 2 - \sin 4\theta \implies \cos 4\theta\,d\theta = -\dfrac{du}{4}$	$-\dfrac{1}{4}\displaystyle\int u^{1/2}\,du$ $= \boxed{-\dfrac{1}{6}(2-\sin 4\theta)^{3/2} + C}$
44	$\displaystyle\int \tan^3 5x\,\sec^2 5x\,dx$	$u = \tan 5x \implies \sec^2 5x\,dx = \dfrac{du}{5}$	$\dfrac{1}{5}\displaystyle\int u^3\,du$ $= \boxed{\dfrac{1}{20}\tan^4 5x + C}$

Ex.	Integral	Choice of u and du	Transformed integral & final answer
45	$\displaystyle\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,dx$	$u=\tan x \implies du=\sec^2 x\,dx$	$\displaystyle\int \frac{du}{\sqrt{1-u^2}}$ $= \boxed{\arcsin(\tan x)+C}$
46	$\displaystyle\int \frac{\sin\theta}{\cos^2\theta+1}\,d\theta$	$u=\cos\theta \implies \sin\theta\,d\theta=-du$	$\displaystyle-\int \frac{du}{u^2+1}$ $= \boxed{-\arctan(\cos\theta)+C}$
47	$\displaystyle\int \sec^3 2x\tan 2x\,dx$	$u=\sec 2x \implies \sec 2x\tan 2x\,dx=\frac{du}{2}$	$\displaystyle\frac{1}{2}\int u^2\,du$ $= \boxed{\frac{1}{6}\sec^3 2x+C}$
48	$\displaystyle\int [\sin(\sin\theta)]\cos\theta\,d\theta$	$u=\sin\theta \implies du=\cos\theta\,d\theta$	$\displaystyle\int \sin u\,du$ $= \boxed{-\cos(\sin\theta)+C}$
49	$\displaystyle\int \frac{dx}{e^x}=\int e^{-x}\,dx$	$u=-x \implies dx=-du$	$\displaystyle-\int e^u\,du$ $= \boxed{-e^{-x}+C}$
50	$\displaystyle\int \sqrt{e^x}\,dx=\int e^{x/2}\,dx$	$u=\frac{x}{2} \implies dx=2\,du$	$\displaystyle 2\int e^u\,du$ $= \boxed{2\sqrt{e^x}+C}$
51	$\displaystyle\int \frac{dx}{\sqrt{x}\,e^{2\sqrt{x}}}$	$u=2\sqrt{x} \implies \frac{dx}{\sqrt{x}}=du$	$\displaystyle\int e^{-u}\,du$ $= \boxed{-e^{-2\sqrt{x}}+C}$
52	$\displaystyle\int \frac{e^{\sqrt{2y+1}}}{\sqrt{2y+1}}\,dy$	$u=\sqrt{2y+1} \implies \frac{dy}{\sqrt{2y+1}}=du$	$\displaystyle\int e^u\,du$ $= \boxed{e^{\sqrt{2y+1}}+C}$
53	$\displaystyle\int \frac{y}{\sqrt{2y+1}}\,dy$	$u=2y+1 \implies y=\frac{u-1}{2},\ dy=\frac{du}{2}$	$\displaystyle\frac{1}{4}\int \left(u^{1/2}-u^{-1/2}\right)du$ $= \boxed{\frac{1}{3}(y-1)\sqrt{2y+1}+C}$
54	$\displaystyle\int x\sqrt{4-x}\,dx$	$u=4-x \implies x=4-u,\ dx=-du$	$\displaystyle\int \left(u^{3/2}-4u^{1/2}\right)du$ $= \boxed{\frac{2}{5}(4-x)^{5/2}-\frac{8}{3}(4-x)^{3/2}+C}$

Ex.	Integral	Choice of u and du	Transformed integral & final answer
55	$\int \sin^3 2\theta \, d\theta$	$\int (1-\cos^2 2\theta)\sin 2\theta \, d\theta, \; u = \cos 2\theta$	$-\frac{1}{2}\int (1-u^2)\, du$ $= \boxed{-\frac{1}{2}\cos 2\theta + \frac{1}{6}\cos^3 2\theta + C}$
56	$\int \sec^4 3\theta \, d\theta$	$\int (1+\tan^2 3\theta)\sec^2 3\theta \, d\theta, \; u = \tan 3\theta$	$\frac{1}{3}\int (1+u^2)\, du$ $= \boxed{\frac{1}{3}\tan 3\theta + \frac{1}{9}\tan^3 3\theta + C}$